

1. A **set with one operation** is called a:
  - a) Group
  - b) Monoid
  - c) Semigroup
  - ☒ d) Algebraic structure
  
2. A **binary operation** on a set is a function that:
  - a) Maps one element to another
  - ☒ b) Combines two elements to produce a third element in the set
  - c) Maps a set to an external element
  - d) Only works for finite sets
  
3. Which of the following properties is NOT necessary for a set with one operation to be a **semigroup**?
  - a) Closure
  - b) Associativity
  - ☒ c) Commutativity
  - d) At least one binary operation
  
4. A **set with one operation** must always satisfy which of the following properties?
  - ☒ a) Closure
  - b) Inverse
  - c) Commutativity
  - d) Identity
  
5. A **semigroup** is a set equipped with a binary operation that is:
  - ☒ a) Associative
  - b) Commutative
  - c) Invertible
  - d) Distributive

6. A **monoid** is a semigroup that also has:
- a) An inverse for every element
  - ☒ b) A unique identity element
  - c) Commutativity
  - d) No closure property
7. If a semigroup contains an **identity element**, it is called a:
- ☒ a) Monoid
  - b) Group
  - c) Field
  - d) Lattice
8. If every element in a monoid has an inverse, the structure becomes a:
- a) Semigroup
  - b) Monoid
  - ☒ c) Group
  - d) Ring
9. A **commutative semigroup** is called:
- ☒ a) Abelian semigroup
  - b) Monoid
  - c) Ring
  - d) Group
10. Which of the following is an example of a **binary operation**?
- ☒ a) Addition of real numbers
  - b) Taking the square root of a number
  - c) Finding the absolute value of a number
  - d) Mapping an integer to its opposite

11. The **identity element** of multiplication in real numbers is:

- a) 0
- ☒ b) 1
- c) -1
- d) 2

12. A binary operation  $*$  is said to be **associative** if:

- ☒ a)  $(a*b)*c=a*(b*c)$  for all  $a,b,c$  in the set
- b)  $a*b=b*a$  for all  $a,b$  in the set
- c)  $a*e=a$  for all  $a$  in the set
- d) Every element has an inverse

13. Which of the following binary operations is NOT **associative**?

- a) Addition of integers
- b) Multiplication of real numbers
- ☒ c) Subtraction of integers
- d) Function composition

14. In **modular arithmetic**, what is  $7+5 \bmod 4$ ?

- ☒ a) 0
- b) 1
- c) 2
- d) 3

15. Which of the following **modular arithmetic properties** is always true for any modulus  $m$ ?

- a)  $(a+b) \bmod m = (a \bmod m + b \bmod m) \bmod m$
- b)  $(a-b) \bmod m = (a \bmod m - b \bmod m) \bmod m$
- c)  $(a \times b) \bmod m = (a \bmod m \times b \bmod m) \bmod m$
- ☒ d) All of the above

16. A **Galois field**  $GF(p)$  is a field with:

- a) An infinite number of elements
- b) A finite number of elements**
- c) No inverse elements
- d) No closure property

17. The number of elements in a **Galois field**  $GF(p^n)$  is:

- a)  $p$
- b)  $p^n$**
- c)  $n^p$
- d)  $p+n$

18. In **modular arithmetic**, the multiplicative inverse of **3 modulo 7** is:

- a) 2
- b) 3
- c) 5**
- d) 6

19. The **basis vectors** of a graph are associated with:

- a) The fundamental cycles of the graph**
- b) The adjacency matrix of the graph
- c) The spanning tree of the graph
- d) The number of isolated vertices in the graph

20. If a **graph has  $n$  vertices and  $m$  edges**, the number of basis vectors in the cycle space is:

- a)  $m-n+1$**
- b)  $m+n-1$
- c)  $n-m+1$
- d)  $m-2n+1$